

ALLIANCE UNIVERSITY

COURSE OF INDEPENDENT STUDY PROPOSALS FOR 2026 ODD SEMESTER

ASAC & ASAE

VII Semester

Prj.N O	School Name	supervisor	Title of the project	Course Name / Transcript Title	Course Code- CIS	Course Code- Course Mapped	Project Description	Prerequisites	Outcomes
1	IT	Prof. Ananthanagu U	Metaverse-Based Smart Campus Navigation System	Metaverse Technology	71T 9269	71T1269	Develop an immersive AR/VR-based campus navigation system that enables users to explore buildings, locate facilities, and receive real-time directions. The system integrates spatial mapping and interactive elements to enhance user experience and aligns with smart city and digital campus initiatives.	1. Basics of AR/VR development 2. Knowledge of 3D modeling and rendering 3. Programming fundamentals (C#/Python) 4. Understanding of spatial mapping concepts 5. Familiarity with immersive interaction design	1. Design and develop an immersive navigation system 2. Apply AR/VR concepts to real-world scenarios 3. Integrate spatial mapping and user interaction 4. Enhance problem-solving in immersive environments 5. Demonstrate industry-relevant XR development skills
2	IT	Dr. Durgadevi P	Design and Develop a Metaverse-Based Smart Campus Experience using AR/VR	Metaverse Technology	71T 9269	71T1269	Develop an immersive virtual representation of a college campus where students, faculty, and visitors can interact in real-time using AR/VR technologies. The system should enable virtual navigation, digital classrooms, event participation, and student services in a 3D metaverse environment.	1. Basic knowledge of programming (C# / JavaScript) 2. Understanding of AR/VR concepts and Metaverse fundamentals 3. Familiarity with Unity or Unreal Engine 4. Basic knowledge of 3D modeling and computer graphics 5. Understanding of database and cloud basics (Azure/AWS/Firebase)	1. Design and develop an immersive 3D virtual campus environment using AR/VR tools 2. Implement avatar-based interaction and real-time communication in a metaverse setting 3. Apply game engine technologies (Unity/Unreal) for interactive system development 4. Integrate cloud services and databases for user management and data handling 5. Demonstrate practical skills in immersive application development and UI/UX design
3	IT	Prof. Ananthanagu U	Immersive Virtual Classroom for Experiential Learning	Metaverse Technology	71T 9269	71T1269	Immersive Virtual Classroom for Experiential Learning	1. Basics of AR/VR and immersive technologies 2. Knowledge of 3D modeling and environment design 3. Programming fundamentals (C#/Python) 4. Understanding of UI/UX for immersive systems 5. Familiarity with networking concepts for multi-user interaction	1. Design and implement an immersive virtual classroom system 2. Apply AR/VR solutions to real-world education challenges 3. Improve accessibility and inclusivity in digital learning 4. Enable remote and rural learners to access quality education 5. Develop practical skills in XR, collaboration systems, and EdTech innovation
4	IT	Dr. Rajesh E	CI/CD Pipeline for Microservices-Based Architecture with Dynamic Scaling	Continuous Integration and Continuous Deployment	71T 9264	71T 1264	Develop a CI/CD pipeline for microservices applications that automates build, test, and deployment while enabling dynamic scaling using container orchestration, ensuring efficient resource utilization and seamless updates across distributed services.	1. Understanding of CI/CD tools (Jenkins, GitHub Actions, GitLab CI) 2. Knowledge of microservices architecture 3. Familiarity with Docker and containerization 4. Basics of Kubernetes or container orchestration 5. Understanding of cloud computing concepts	1. Design and implement CI/CD pipelines for microservices 2. Automate independent build and deployment processes 3. Apply containerization and orchestration techniques 4. Implement dynamic scaling for distributed services 5. Analyze system performance and scalability in cloud environments
5	IT	Dr. Ummadi Janardhan	CI/CD Pipeline for Smart Healthcare Monitoring Systems	Continuous Integration and Continuous	71T 9264	71T 1264	Develop a CI/CD pipeline for continuous deployment of healthcare monitoring applications with real-time updates.	1. DevOps fundamentals 2. CI/CD tools (Jenkins/GitHub Actions) 3. Cloud basics 4. Git version control	1. Automate healthcare deployments 2. Improve system reliability 3. Enable continuous updates 4. Integrate monitoring tools 5. Apply DevOps in healthcare
6	IT	Dr. Rajesh E	Green CI/CD Framework for Energy-Efficient Software Deployment and Pipeline Optimization	Continuous Integration and Continuous	71T 9264	71T 1264	Design a CI/CD pipeline that monitors and optimizes energy and resource consumption during build, test, and deployment stages, using intelligent scheduling and performance analysis to reduce carbon footprint and improve efficiency.	1. Basics of Software Development Life Cycle (SDLC) 2. Understanding of CI/CD concepts and tools (e.g., Jenkins, GitHub Actions) 3. Knowledge of Docker and containerization 4. Basic understanding of cloud computing 5. Familiarity with scripting (Python/Shell) 6. Basic knowledge of system performance metrics (CPU, memory usage)	1. Design and implement an energy-aware CI/CD pipeline 2. Analyze and optimize resource utilization across pipeline stages 3. Apply scheduling strategies to improve efficiency 4. Develop monitoring dashboards for energy and performance metrics 5. Evaluate sustainability impact of CI/CD processes
7	IT	Dr Vinita Nair	Design, Performance Evaluation, and Optimization of CI/CD Pipelines in Agile Environments	Continuous Integration and Continuous	71T 9264	71T 1264	This project focuses on designing, evaluating, and optimizing CI/CD pipelines in Agile environments. It aims to improve software delivery speed, reliability, and automation efficiency using modern DevOps tools and practices	1. Basics of Software Engineering and Agile Methodologies 2. Knowledge of Version Control Systems (e.g., Git) 3. Understanding of DevOps Concepts 4. Familiarity with CI/CD Tools (Jenkins, GitHub Actions, GitLab CI) 5. Basic Scripting Skills (Python/Bash)	1. Ability to design efficient CI/CD pipelines for Agile workflows 2. Skills in integrating automated testing and deployment processes 3. Capability to evaluate pipeline performance using metrics 4. Knowledge of optimization techniques for faster and reliable delivery 5. Hands-on experience with industry-standard DevOps tools and practices
8	IT	Dr Jayabhaduri R	Data Analytics for Continuous Integration and Continuous Deployment Performance Optimization	Continuous Integration and Continuous	71T 9264	71T 1264	Develop an analytics framework that collects and analyzes CI/CD pipeline data to predict build failures, optimize deployment frequency, reduce lead time, and improve software delivery performance using machine learning and visualization techniques.	1. Python Programming 2. Data Analytics 3. Machine Learning Fundamentals 4. Software Engineering 5. Git and Version Control 6. Basic DevOps Concepts	1. Collect and preprocess CI/CD pipeline data. 2. Develop predictive models for build and deployment failures. 3. Analyze key DevOps metrics such as lead time and deployment frequency. 4. Create dashboards for monitoring CI/CD performance. 5. Recommend data-driven improvements for software delivery pipelines.
9	IT	Prof. Kusuma J	Docker-Based Multi-Environment Deployment System	Continuous Integration and Continuous	71T 9264	71T 1264	This project focuses on containerizing applications using Docker and deploying them across multiple environments such as development and testing. It ensures consistency, scalability, and reliable deployments while reducing environment-related issues in software delivery	1. Basic understanding of operating systems and Linux 2. Knowledge of Docker and containerization basics 3. Familiarity with web applications 4. Understanding of networking fundamentals 5. Basic scripting knowledge (Shell/Python) 6. Awareness of DevOps practices	1. Ability to containerize applications using Docker 2. Understanding of multi-environment deployment 3. Knowledge of environment consistency and scalability 4. Hands-on experience with deployment automation 5. Ability to manage and run applications in containers
10	IT	Prof.Kusuma J	Automated CI/CD Pipeline for Web Application Deployment	Continuous Integration and Continuous	71T 9264	71T 1264	This project develops an automated CI/CD pipeline to build, test, and deploy a web application using DevOps tools. It ensures faster delivery, reduces manual errors, and improves software quality through continuous integration and deployment processes.	1. Basic knowledge of software development lifecycle 2. Understanding of Git and version control 3. Familiarity with programming (Python/Node.js/Java) 4. Basic knowledge of Linux commands 5. Introduction to DevOps concepts 6. Awareness of tools like Jenkins or GitHub Actions	1. Ability to design and implement CI/CD pipelines 2. Understanding of automated testing and deployment 3. Experience with DevOps tools and workflows 4. Improved knowledge of software delivery lifecycle 5. Ability to integrate code, build, test, and deploy automatically
11	IT	Dr. U. Janardhan Reddy	Multi-Cloud CI/CD System for Smart Education Platforms	Continuous Integration and Continuous	71T 9264	71T 1264	Develop a CI/CD pipeline supporting deployment of education platforms across multiple cloud environments.	1. Cloud computing basics 2. Kubernetes & Docker 3. CI/CD tools 4. Terraform basics 5. Networking concepts	1. Deploy across multi-cloud 2. Improve scalability 3. Automate infrastructure 4. Enable cloud portability 5. Optimize deployment pipelines
12	IT	Prof. Kusuma J	Docker-Based Multi-Environment Deployment System	Continuous Integration and Continuous	71T 9264	71T 1264	This project focuses on integrating automated security checks into a CI/CD pipeline. The system performs source-code analysis, dependency vulnerability scanning, secret detection, and security validation during the software delivery process. It helps identify security issues early and prevents vulnerable code from progressing to deployment	1. Basic knowledge of software development lifecycle 2. Understanding of Git and version control 3. Familiarity with Python/Java/JavaScript 4. Basic Linux commands 5. Basic knowledge of web application security 6. Introduction to CI/CD concepts 7. Basic understanding of software vulnerabilities	1. Ability to integrate security checks into CI/CD pipelines 2. Understanding of DevSecOps principles 3. Ability to perform automated vulnerability and dependency scanning 4. Ability to identify security issues during software development 5. Understanding of secure software delivery practices 6. Hands-on experience with security automation tools

13	CSE - Data Analytics	Dr. S. Iwin Thanakumar Joseph	AgroVisionNet: Deep Neural Network Framework for Precision Agriculture and Crop Disease Intelligence	Deep Learning and Neural Network	7CS 9014	7CSI014	AgroVisionNet leverages Convolutional Neural Networks (CNNs), Vision Transformers (ViTs), and predictive analytics to identify crop diseases, assess plant health, forecast yields, and provide actionable farming insights. The platform supports precision agriculture through automated monitoring, intelligent diagnosis, and data-driven decision making.	Python Programming Data Structures and Algorithms Machine Learning Deep Learning Computer Vision Convolutional Neural Networks (CNNs) Vision Transformers (ViTs) Image Processing Database Management Systems Cloud Computing	Detect and classify crop diseases with high accuracy. Monitor crop health using image-based intelligence. Generate precision farming recommendations. Predict crop yield and potential productivity risks. Optimize the use of water, fertilizers, and pesticides. Improve agricultural efficiency, sustainability, and farm profitability through AI-driven insights.
14	CSE - Data Analytics	Dr. S. Iwin Thanakumar Joseph	KrishiGenAI: Intelligent Agricultural Advisory and Crop Planning Assistant	Generative AI	7CS 9018	7CSI018	The system utilizes Generative AI, machine learning, weather data, soil information, and agricultural knowledge bases to generate customized farming recommendations. It assists farmers in crop selection, pest management, resource optimization, and market planning, enabling informed decision-making and enhanced agricultural productivity.	Python Programming Machine Learning Deep Learning Generative AI and LLMs Natural Language Processing (NLP) Data Analytics Database Management Systems API Integration Cloud Computing Basic Agricultural Domain Knowledge	Generate personalized crop and cultivation recommendations. Predict and identify crop diseases and pest infestations. Optimize irrigation and fertilizer usage. Provide weather-based agricultural advisories. Improve crop yield and farm profitability. Support sustainable and data-driven farming practices.
15	CSE - Data Analytics	Dr. S. Iwin Thanakumar Joseph	National Knowledge Graph and Generative AI Assistant for Government Schemes	Generative AI	7CS 9018	7CSI018	The platform combines Knowledge Graphs, Large Language Models (LLMs), and Retrieval-Augmented Generation (RAG) to deliver trustworthy and context-aware guidance on government schemes. It enables intelligent scheme discovery, eligibility assessment, document assistance, and multilingual citizen support while ensuring transparency and reducing misinformation.	Python Programming Data Structures and Algorithms Machine Learning Deep Learning Generative AI and LLMs Knowledge Graphs and Graph Databases Natural Language Processing (NLP) Retrieval-Augmented Generation (RAG) Database Management Systems Cloud Computing	Provide accurate and explainable information on government schemes. Recommend schemes based on citizen profiles and eligibility. Enable multilingual citizen interaction and support. Reduce information gaps and improve scheme accessibility. Enhance transparency and efficiency in public service delivery. Demonstrate the integration of Knowledge Graphs with Generative AI for governance applications.
16	CSE - Data Analytics	Dr. S. Iwin Thanakumar Joseph	BharatGPT: Multilingual Generative AI Platform for Citizen-Centric Governance	Generative AI	7CS 9018	7CSI018	BharatGPT is a multilingual Generative AI platform that enables citizens to access government services, schemes, grievance redressal, and policy information through natural language interactions. It leverages Large Language Models, Retrieval-Augmented Generation (RAG), and regional language support to enhance transparency, accessibility, and citizen engagement.	Python Programming Machine Learning Fundamentals Deep Learning Basics Natural Language Processing (NLP) Generative AI and Large Language Models (LLMs) Retrieval-Augmented Generation (RAG) Database Management Systems Web Development (FastAPI/Flask, React)	Develop a multilingual AI-powered governance assistant. Enable seamless access to government schemes and services. Automate citizen query handling and grievance support. Improve accessibility for regional language users. Enhance transparency, efficiency, and inclusivity in public service delivery. Demonstrate practical implementation of Generative AI for Digital India initiatives.
17	CSE - Data Analytics	Prof. Lalli	Autonomous Generative AI System for Supply Chain Risk Prediction and Resilience Planning	Generative AI	7CS 9018	7CSI018	The system integrates Large Language Models (LLMs), predictive analytics, digital twins, and Retrieval-Augmented Generation (RAG) to identify supply chain vulnerabilities, forecast disruptions, and autonomously generate mitigation plans. It supports proactive decision-making, risk management, and resilient supply chain operations across diverse industries.	Python Programming Data Structures and Algorithms Machine Learning Deep Learning Generative AI and LLMs Retrieval-Augmented Generation (RAG) Predictive Analytics Database Management Systems Cloud Computing Supply Chain Management Fundamentals	Predict supply chain risks and disruptions in advance. Generate intelligent resilience and mitigation strategies. Optimize inventory, sourcing, and logistics planning. Improve supply chain visibility and operational efficiency. Support data-driven decision-making and business continuity. Enhance organizational resilience against market, geopolitical, and environmental uncertainties.
18	CSE - Data Analytics	Prof. Lalli	GreenGen: Generative AI Platform for Carbon Footprint Optimization and Sustainable Development Planning	Generative AI	7CS 9018	7CSI018	GreenGen leverages Large Language Models (LLMs), predictive analytics, and sustainability data to assess carbon emissions, identify inefficiencies, and generate personalized decarbonization roadmaps. The platform supports ESG reporting, renewable energy planning, and sustainable development initiatives through AI-driven insights and decision support.	Python Programming	Calculate and analyze carbon footprints across operations.
19	CSE - Data Analytics	Dr. S. Iwin Thanakumar Joseph	AI-Driven Cyber Threat Intelligence and Incident Report Generation System	Generative AI	7CS 9018	7CSI018	The system leverages Large Language Models, machine learning, and threat intelligence frameworks to monitor security events, detect anomalies, correlate attack patterns, and generate comprehensive incident reports. It assists security teams in threat hunting, risk assessment, incident response, and cybersecurity compliance management.	Python Programming	Detect and analyze cyber threats in real time.
20	CSE - Data Analytics	Dr. S. Iwin Thanakumar Joseph	Generative AI-Driven Disaster Intelligence and Emergency Response System	Generative AI	7CS 9018	7CSI018	The system leverages Generative AI, Large Language Models, and predictive analytics to provide real-time disaster insights, automate emergency planning, optimize rescue operations, and generate actionable response strategies. It assists authorities in minimizing casualties, improving resource utilization, and enhancing disaster preparedness and recovery efforts.	Python Programming Machine Learning Deep Learning Generative AI and LLMs Natural Language Processing (NLP) Data Analytics Geospatial Information Systems (GIS) Cloud Computing Database Management Systems API Integration	Generate real-time disaster intelligence reports. Automate evacuation and emergency response planning. Optimize resource allocation during disasters. Improve situational awareness for authorities. Reduce disaster response time and operational costs. Enhance public safety and disaster resilience through AI-driven decision support.
21	CSE - Data Analytics	Prof. Lalli	SkillForge AI: Personalized Career Roadmap Generation Using Agentic AI and Labor Market Intelligence	Generative AI	7CS 9018	7CSI018	SkillForge AI combines Generative AI, Agentic AI, labor market analytics, and Large Language Models (LLMs) to assess user competencies, identify skill gaps, predict emerging career opportunities, and generate adaptive learning and career pathways aligned with industry demands and future workforce requirements.	Python Programming Data Structures and Algorithms Machine Learning Deep Learning Generative AI and LLMs Agentic AI and Multi-Agent Systems Natural Language Processing (NLP) Data Analytics and Visualization Database Management Systems Cloud Computing	Generate personalized career roadmaps and learning plans. Identify skill gaps and recommend upskilling opportunities. Analyze labor market trends and future job demands. Provide intelligent job and career recommendations. Support data-driven career planning and workforce readiness. Enhance employability through AI-powered career guidance.
22	CSE - Data Analytics	Dr. S. Iwin Thanakumar Joseph	AquaGenAI: Intelligent Water Resource Management and Conservation Planning Platform	Generative AI	7CS 9018	7CSI018	AquaGenAI integrates Generative AI, predictive analytics, IoT sensor data, GIS, and hydrological intelligence to monitor water resources, forecast demand, detect inefficiencies, and generate conservation recommendations. The platform enables data-driven planning, equitable distribution, and sustainable utilization of water resources across urban and rural regions.	Python Programming Data Structures and Algorithms Machine Learning Deep Learning Generative AI and LLMs Internet of Things (IoT) Geographic Information Systems (GIS) Data Analytics and Visualization Database Management Systems Cloud Computing	Forecast water demand and availability using AI models. Generate intelligent water conservation and allocation strategies. Detect water wastage, leakages, and resource inefficiencies. Support drought preparedness and sustainable resource planning. Improve decision-making for water authorities and policymakers. Promote efficient and equitable water resource management through AI-driven insights.

23	CSE - Data Analytics	Dr. S. Iwin Thanakumar Joseph	NeuroRAG: Brain-Inspired Retrieval-Augmented Generation System for Explainable Decision Intelligence	Generative AI	7CS 9018	7CS1018	NeuroRAG combines Large Language Models (LLMs), Retrieval-Augmented Generation, knowledge graphs, and cognitive-inspired architectures to retrieve relevant information, perform contextual reasoning, and generate explainable recommendations. The system enhances transparency, reduces AI hallucinations, and provides trustworthy decision intelligence through evidence-backed outputs and reasoning traces.	Python Programming Data Structures and Algorithms Machine Learning Deep Learning Generative AI and LLMs Retrieval-Augmented Generation (RAG) Knowledge Graphs Natural Language Processing (NLP) Explainable AI (XAI) Database Management Systems	Develop an explainable Retrieval-Augmented Generation framework. Generate evidence-backed recommendations and decisions. Reduce hallucinations through intelligent knowledge retrieval. Improve transparency and trustworthiness of AI systems. Enable reasoning traceability for decision support applications. Demonstrate brain-inspired AI architectures for intelligent and explainable decision-making.
24	CSE - Data Analytics	Dr. S. Iwin Thanakumar Joseph	Smart Urban Planning Assistant Using Generative AI and Digital Twin Technology	Generative AI	7CS 9018	7CS1018	The system combines Generative AI, Digital Twins, geospatial analytics, and real-time urban data to model city environments, predict infrastructure demands, evaluate planning scenarios, and support informed decision-making. It assists urban planners in optimizing resource allocation, sustainability initiatives, and smart city development strategies.	Python Programming Machine Learning Deep Learning Generative AI and LLMs Digital Twin Technology Geographic Information Systems (GIS) Data Analytics and Visualization Database Management Systems Cloud Computing Smart Cities and Urban Planning Fundamentals	Create a digital twin of urban infrastructure. Generate intelligent recommendations for city planning. Simulate and evaluate urban development scenarios. Optimize transportation, utilities, and resource management. Support sustainable and data-driven decision-making. Enhance smart city planning, resilience, and citizen well-being.
25	CSE - Data Analytics	Dr. S. Iwin Thanakumar Joseph	Generative AI Framework for Environmental Impact Assessment and Sustainability Reporting	Generative AI	7CS 9018	7CS1018	The platform leverages Generative AI, machine learning, and environmental analytics to assess ecological impacts, evaluate sustainability metrics, and generate comprehensive EIA and ESG reports. It provides actionable recommendations for resource optimization, regulatory compliance, and sustainable development while reducing manual effort and reporting time.	Python Programming Machine Learning Deep Learning Generative AI and LLMs Data Analytics and Visualization Environmental Science Fundamentals ESG and Sustainability Reporting Concepts Natural Language Processing (NLP) Database Management Systems Cloud Computing	Automate Environmental Impact Assessment (EIA) processes. Generate ESG and sustainability reports automatically. Predict environmental risks and ecological impacts. Improve regulatory compliance and transparency. Support sustainable decision-making and resource management. Reduce reporting time and operational costs through AI-driven automation.
26	CSE - Data Analytics	Prof. Lalli	HeritageGPT: Generative AI Platform for Cultural Heritage Preservation and Tourism	Generative AI	7CS 9018	7CS1018	HeritageGPT integrates Large Language Models, Retrieval-Augmented Generation (RAG), computer vision, and geospatial technologies to generate historical narratives, virtual heritage tours, multilingual cultural insights, and tourism recommendations. The platform supports heritage preservation, visitor engagement, and sustainable tourism through AI-driven knowledge dissemination.	Python Programming Machine Learning Deep Learning Generative AI and LLMs Natural Language Processing (NLP) Computer Vision Retrieval-Augmented Generation (RAG) Database Management Systems Geographic Information Systems (GIS) Cloud Computing	Digitize and preserve cultural heritage knowledge. Generate multilingual historical narratives and guided tours. Provide personalized tourism recommendations. Enhance visitor engagement through AI-powered experiences. Promote cultural awareness and heritage conservation. Support sustainable tourism and digital heritage initiatives.
27	CSE - Data Analytics	Dr. S. Iwin Thanakumar Joseph	InfraMind: Generative AI System for Predictive Infrastructure Maintenance and Smart Asset Management	Generative AI	7CS 9018	7CS1018	InfraMind integrates Generative AI, predictive analytics, IoT data, and Digital Twin technologies to monitor infrastructure assets, forecast maintenance requirements, and generate intelligent maintenance plans. The platform supports data-driven asset management, risk mitigation, lifecycle optimization, and operational efficiency across transportation, energy, and public infrastructure sectors.	Python Programming Data Structures and Algorithms Machine Learning Deep Learning Generative AI and LLMs Internet of Things (IoT) Digital Twin Technology Predictive Analytics Database Management Systems Cloud Computing	Predict infrastructure failures and maintenance requirements. Generate AI-driven maintenance schedules and recommendations. Improve asset reliability and operational efficiency. Reduce maintenance costs and unplanned downtime. Enable data-driven smart asset management. Enhance the sustainability and resilience of critical infrastructure systems.
28	CSE - Data Analytics	Dr. S. Iwin Thanakumar Joseph	GeoNarrator AI: Generative Geospatial Intelligence Platform for Strategic Infrastructure Development	Generative AI	7CS 9018	7CS1018	GeoNarrator AI is a Generative AI-powered geospatial intelligence platform that analyzes satellite imagery, geographic data, and infrastructure information to support strategic planning and development. The system generates location-based insights, risk assessments, and infrastructure recommendations for informed decision-making and sustainable urban growth.	Programming Fundamentals using Python Database Management Systems (DBMS) Data Analytics and Visualization Geographic Information Systems (GIS) and Remote Sensing Basics Machine Learning, Artificial Intelligence, and Generative AI Fundamentals	Design and develop a Generative AI-based geospatial intelligence platform for infrastructure planning and development. Integrate geospatial datasets, satellite imagery, and infrastructure information for spatial analysis and decision support. Apply AI and geospatial analytics techniques to identify suitable locations, infrastructure gaps, risks, and development opportunities. Develop interactive visualization dashboards and natural language interfaces for geospatial data exploration and reporting. Evaluate the effectiveness of the platform in supporting strategic infrastructure planning, resource optimization, and sustainable development initiatives.
29	CSE - Data Analytics	Dr. S. Iwin Thanakumar Joseph	Project ANVESHAN: Agentic Generative AI for Autonomous Knowledge Discovery from Government Data	Generative AI	7CS 9018	7CS1018	Project ANVESHAN develops an Agentic Generative AI platform that autonomously analyzes government datasets to discover hidden patterns, generate insights, identify trends, and support evidence-based decision making. The system leverages AI agents, knowledge graphs, and large language models to transform complex public data into actionable intelligence.	Programming Fundamentals using Python Database Management Systems and Data Warehousing Data Analytics and Visualization Machine Learning and Artificial Intelligence Fundamentals Basic Understanding of Generative AI, Large Language Models (LLMs), and Agentic AI Systems	Design and develop an agentic Generative AI framework for autonomous analysis of government and public datasets. Implement AI agents capable of data retrieval, preprocessing, pattern discovery, insight generation, and report creation. Apply Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) techniques to generate explainable insights from structured and unstructured government data. Develop interactive dashboards and natural language query interfaces for exploring public datasets and policy-related information. Evaluate the effectiveness of the proposed system in supporting data-driven governance, policy analysis, and public sector decision-making.
30	CSE General	Dr. Sridhar D	Digital Twin for Personalized Healthcare using Generative AI	Generative AI	7CS 9018	7CS1018	Develop an AI-powered Digital Twin platform that integrates Electronic Health Records (EHR), wearable sensor data, and medical imaging to create personalized virtual patient models. Using Generative AI and Retrieval-Augmented Generation (RAG), the system predicts health risks, generates clinical insights, and supports informed decision-making for improved patient care and healthcare outcomes.	1. Python Programming Fundamentals 2. Machine Learning and Deep Learning Basics 3. Database Management Systems (SQL/NoSQL) 4. Basics of Generative AI and Large Language Models (LLMs) 5. Healthcare Data Analytics and REST API Fundamentals	1. Develop a Digital Twin healthcare system. 2. Integrate Generative AI for personalized health insights. 3. Implement RAG for intelligent healthcare information retrieval. 4. Deploy a secure AI-powered healthcare application.
31	CSE General	Dr. Rashmi Mothkur	DiagnoGen ALA Generative AI and Bio-Inspired Optimization Framework for Accurate Disease Diagnosis and Healthcare Intelligence	Generative AI	7CS 9018	7CS1018	This project investigates the use of Generative AI and bio-inspired optimization algorithms for disease detection from medical images. The project focuses on enhancing image quality, extracting relevant features, and improving diagnostic accuracy through intelligent AI-driven healthcare solutions	1. Python Programming Fundamentals 2. Basics of Medical Image Processing 3. Machine Learning Fundamentals 4. Deep Learning Concepts 5. Introduction to Generative AI 6. Knowledge of Bio-Inspired Optimization Techniques	1. Design and develop a Generative AI-based system for disease diagnosis from medical images. 2. Implement image enhancement and feature extraction techniques for improved diagnostic accuracy. 3. Apply bio-inspired optimization algorithms for efficient feature selection and model tuning. 4. Develop a disease classification module using deep learning and medical image analysis. 5. Provide explainable diagnostic insights and clinical decision support. 6. Build a user-friendly interface for image upload and diagnosis reporting. 7. Evaluate system performance using accuracy, precision, recall, and F1-score metrics.

32	CSE General	Dr. Rajasckar	LearnGen: Adaptive Personalized Learning Platform Using Generative AI	Generative AI	7CS 9018	7CSI018	LearnGen is a Generative AI-powered adaptive learning platform that personalizes educational content, assessments, and learning pathways based on student performance and learning preferences. The system provides intelligent tutoring, automated feedback, and real-time progress tracking to enhance learning outcomes and engagement.	1.Programming Fundamentals using Python 2.Data Structures and Algorithms 3.Database Management Systems (DBMS) 4.Web Application Development 5.Machine Learning and Artificial Intelligence Fundamentals 6.Basic Understanding of Generative AI and Large Language Models (LLMs)	1.Design and develop an adaptive learning platform that delivers personalized educational content using Generative AI. 2.Implement AI-driven learning path generation based on student performance, learning style, and competency levels. 3.Develop intelligent assessment and feedback mechanisms for automated quiz generation and performance evaluation. 4.Integrate Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) to provide contextual tutoring and doubt resolution. 5.Build an interactive web/mobile application supporting personalized learning, progress tracking, and performance analytics. 6.Evaluate the effectiveness of the platform in improving student engagement, knowledge retention, and learning outcomes.
33	CSE General	Dr. Oswalt Manoj	AgroSageGPT: A Multilingual Generative AI Digital Agronomist for Sustainable Farming	Generative AI	7CS 9018	7CSI018	AgroSageGPT is a multilingual Generative AI-powered digital agronomist that assists farmers in crop planning, disease diagnosis, irrigation management, and market decision-making. By integrating AI, computer vision, weather intelligence, and agricultural knowledge, it delivers personalized, explainable, and sustainable farming recommendations through text, voice, and image-based interactions.	1.Programming Fundamentals using Python 2.Data Structures and Algorithms 3.Database Management Systems (DBMS) 4.Machine Learning and Artificial Intelligence Fundamentals 5.Web Application Development (Frontend and Backend Basics) 6.Basic Understanding of Generative AI, NLP, and Large Language Models (LLMs)	1.Design and develop a multilingual Generative AI-based digital agronomist for precision and sustainable farming applications. 2.Implement Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) to provide context-aware agricultural recommendations related to crops, soil, irrigation, and fertilizers. 3.Develop a computer vision module for crop disease and pest detection using image analysis techniques. 4.Integrate weather, market, and agricultural knowledge sources to generate personalized, explainable, and location-specific farming advisories. 5.Build an interactive web/mobile platform supporting text, voice, and image-based farmer interactions. 6.Evaluate the effectiveness of the proposed system in improving agricultural productivity, resource optimization, and sustainable farming practices.
34	CSE General	Ms. Pragna	Hallucination Detection and Correction Framework for Large Language Models using Retrieval-Augmented Generation (RAG)	Generative AI	7CS 9018	7CSI018	Large Language Models may generate factually incorrect or fabricated responses due to insufficient context. This project develops a Retrieval-Augmented Generation (RAG)-based framework that detects unsupported content, verifies responses using trusted knowledge sources, and regenerates factually accurate answers, improving the reliability and trustworthiness of Generative AI systems. preparedness, response coordination, and recovery through intelligent decision support	1.Python Programming Fundamentals 2.Machine Learning and Deep Learning Basics 3.Data Structures and Algorithms 4.Natural Language Processing (NLP) 5.Generative AI and Large Language Models (LLMs)	1. Design and implement a RAG-based framework for hallucination detection. 2. Retrieve relevant knowledge from trusted document repositories using vector similarity search. 3. Detect unsupported or hallucinated statements in LLM-generated responses. 4. Develop a correction mechanism that regenerates responses using verified evidence. 5. Evaluate the framework using factual accuracy, hallucination rate, answer relevance, and response quality metrics. 6. Compare the proposed framework with a baseline LLM and a standard RAG system to demonstrate performance improvements.
35	CSE General	Dr. Karpagalakshmi	CivicPulse AI: A Multilingual Generative AI Platform for Smart Civic Issue Reporting, Prioritization, and Citizen-Centric Urban Governance	Generative AI	7CS 9018	7CSI018	CivicPulse AI is a multilingual civic grievance management platform that uses Generative AI to collect, analyze, prioritize, and route citizen complaints. It enhances communication between citizens and government agencies, enabling faster issue resolution, improved transparency, and efficient urban governance.	1. Python Programming 2. Machine Learning Fundamentals 3. Generative AI and Large Language Models (LLMs) 4. Natural Language Processing (NLP) 5. Computer Vision Basics	1. Develop a multilingual AI-powered civic complaint management platform. 2. Enable citizens to report issues through voice, text, and image inputs. 3. Automate complaint classification and prioritization using Generative AI. 4. Improve response time by intelligently routing complaints to appropriate authorities. 5. Provide real-time complaint tracking and status updates. 6. Enhance citizen engagement and participation in urban governance.
36	CSE-AI/ML	Dr. Nishanth Tripathi	Automated Student Performance Prediction	AUTO ML	7CS 9281	7CSI281	This project develops a machine learning-based system to predict student academic performance using historical academic records, attendance, assessment scores, and behavioral indicators. The system assists educators in identifying at-risk students early and enables data-driven interventions to improve learning outcomes and academic success.	1.Basics of Python Programming 2.Fundamentals of Machine Learning 3. Data Preprocessing Techniques 4. Knowledge of Statistics and Probability 5. Understanding of Classification and Regression Models 6. Experience with Pandas, NumPy, and Scikit-learn 7. Basic Data Visualization Concepts	1.Design and develop a predictive analytics system for student performance forecasting. 2.Apply machine learning algorithms to educational datasets. 3. Perform data preprocessing, feature engineering, and model evaluation. 4. Compare multiple prediction models and identify the best-performing approach
37	CSE-AI/ML	Dr. Nishanth Tripathi	Employee Attrition Prediction using AutoML	AUTO ML	7CS 9281	7CSI281	Develop an AutoML-based predictive system to identify employees who are likely to leave an organization. The project analyzes workforce data such as job role, performance, salary, and work experience to support proactive employee retention strategies.	1.Basic Python Programming 2.Fundamentals of Machine Learning 3. Basic Data Analysis and Visualization	1. Build and evaluate an AutoML model for employee attrition prediction. 2. Identify key factors influencing employee turnover. 3. Generate data-driven insights to support retention and workforce planning decisions.
38	CSE-AI/ML	Dr. M Srinivas	House Price Prediction using AutoML	AUTO ML	7CS 9281	7CSI281	Develop an AutoML-based predictive system to estimate house prices using property features such as location, area, number of rooms, amenities, and market trends. The project enables accurate real estate valuation and supports informed property investment decisions.	1. Basic Python Programming 2. Fundamentals of Machine Learning 3. Basic Data Analysis and Visualization	1.Build and evaluate an AutoML model for house price prediction. 2. Identify key property features influencing housing prices. 3. Generate accurate price estimates to support real estate decision-making
39	CSE-AI/ML	Dr. M Srinivas	Loan Approval Prediction System	AUTO ML	7CS 9281	7CSI281	Develop an AutoML-based intelligent system to predict loan approval decisions using applicant information such as income, employment status, credit history, and financial records. The system assists financial institutions in making faster, data-driven, and consistent lending decisions	1. Basic Python Programming 2. Fundamentals of Machine Learning 3. Basic Data Analysis and Visualization	1. Build and evaluate an AutoML model for loan approval prediction. 2. Identify key financial factors influencing loan decisions. 3. Generate reliable predictions to support automated lending processes.
40	CSE-AI/ML	Dr. M Srinivas	Automated Missing Value Imputation Framework	AUTO ML	7CS 9281	7CSI281	Develop an intelligent AutoML-based framework to automatically identify and impute missing values in datasets using statistical, machine learning, and data-driven techniques. The framework improves data quality and enhances the reliability of predictive analytics and decision-making processes.	1. Basic Python Programming 2. Fundamentals of Machine Learning 3. Data Preprocessing and Data Analysis Concepts	1. Develop an automated framework for detecting and imputing missing data. 2. Compare multiple imputation techniques and identify the most effective approach. 3. Improve dataset quality to enhance machine learning model performance.
41	CSE-AI/ML	Dr. Ashok Sarabu	Smart Feature Selection Engine	AUTO ML	7CS 9281	7CSI281	Develop an intelligent feature selection engine that automatically identifies the most relevant attributes from a dataset using statistical and machine learning techniques. The system reduces data dimensionality, improves model efficiency, and enhances predictive performance.	1. Basic Python Programming 2. Fundamentals of Machine Learning 3. Data Preprocessing and Feature Engineering Concept	1. Develop an automated system for selecting relevant features from datasets. 2. Evaluate and compare multiple feature selection techniques. 3. Improve model accuracy, interpretability, and computational efficiency through optimal feature selection
42	CSE-AI/ML	Dr. Ashok Sarabu	Customer Segmentation using AutoML	AUTO ML	7CS 9281	7CSI281	Develop an AutoML-based system to segment customers using purchasing behavior, demographics, spending patterns, and engagement data. The system helps businesses identify customer groups and design targeted marketing strategies	1. Basic Python Programming 2. Machine Learning Fundamentals 3. Data Analysis and Visualization Basics	1. Build an AutoML-based customer segmentation model. 2. Identify meaningful customer groups from business data. 3. Support targeted marketing and personalized business decisions
43	CSE-AI/ML	Dr. Ashok Sarabu	Auto Data Cleaning Assistant	AUTO ML	7CS 9281	7CSI281	Develop an intelligent data cleaning assistant that automatically detects and handles missing values, duplicates, inconsistencies, and outliers in datasets. The system improves data quality and prepares datasets for reliable analytics and machine learning applications	1. Basic Python Programming 2. Data Preprocessing Fundamentals 3. Basic Machine Learning Concepts	1. Develop an automated framework for cleaning and preprocessing datasets. 2. Detect and resolve common data quality issues efficiently. 3. Generate high-quality datasets for improved analytical and predictive performance
44	CSE-AI/ML	Dr. Nishanth Tripathi	Hyperparameter Optimization Comparison Framework	AUTO ML	7CS 9281	7CSI281	This project develops an AutoML framework that compares multiple hyperparameter optimization techniques such as Grid Search, Random Search, and Bayesian Optimization. The platform evaluates their impact on model performance, training time, and computational efficiency, enabling informed selection of optimization strategies for machine learning applications.	1. Programming Fundamentals using Python 2. Machine Learning Fundamentals 3. Statistics and Probability Basics 4. Data Analytics and Visualization Fundamentals 5. Basic Understanding of AutoML and Model Evaluation Techniques	1. Design and develop a comparative framework for evaluating multiple hyperparameter optimization techniques in machine learning. 2. Implement and benchmark optimization methods such as Grid Search, Random Search, and Bayesian Optimization across various datasets and models. 3. Analyze the impact of optimization techniques on model accuracy, training time, resource utilization, and computational efficiency. 4. Generate comparative reports, visualizations, and recommendations to support effective hyperparameter tuning and model optimization decisions.

45	CSE-AI/ML	Dr. Irwin ThanaKumar	Auto Classifier Benchmarking System	AUTO ML	7CS 9281	7CSI281	This project develops an automated benchmarking platform that trains, evaluates, compares, and ranks machine learning classification models across diverse datasets. The system provides performance insights and model recommendations, enabling users to efficiently identify the most suitable classification algorithms for predictive analytics tasks.	Programming Fundamentals using Python Database Management Systems (DBMS) Machine Learning Fundamentals Data Analytics and Visualization Basics Statistics and Performance Evaluation Metrics	1. Design and develop an automated benchmarking platform for training, evaluating, and comparing multiple machine learning classification models. 2. Implement performance evaluation using metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and execution time. 3. Develop a ranking and recommendation engine that identifies the most suitable classification models for different datasets. 4. Generate interactive reports and visualizations to support data-driven model selection and comparative performance analysis.
46	CSE-AI/ML	Dr. M Srinivas	IoT Sensor Failure Prediction	AUTO ML	7CS 9281	7CSI281	This project develops an AutoML-based predictive maintenance system that analyzes IoT sensor data to forecast equipment failures before they occur. The platform enables proactive maintenance, reduces operational downtime, improves asset reliability, and supports data-driven decision-making in industrial and smart manufacturing environments.	1. Programming Fundamentals using Python 2. Database Management Systems (DBMS) 3. Internet of Things (IoT) Fundamentals 4. Machine Learning and AutoML Basics 5. Data Analytics and Visualization Fundamentals	1. Design and develop an AutoML-based predictive maintenance system for forecasting equipment and sensor failures using IoT data. 2. Implement automated data preprocessing, feature selection, and model generation techniques to identify failure patterns and anomalies. 3. Analyze real-time and historical sensor data to predict potential equipment breakdowns and maintenance requirements. 4. Develop monitoring dashboards and alert mechanisms to support proactive maintenance, reduce downtime, and improve operational reliability.
47	CSE-AI/ML	Dr. Ashok Sarabu	AutoML Model Recommendation System	AUTO ML	7CS 9281	7CSI281	This project develops an intelligent AutoML recommendation system that analyzes dataset characteristics and automatically suggests suitable machine learning algorithms and workflows. The platform assists users in selecting optimal models, improving predictive performance, reducing experimentation time, and simplifying the machine learning development process.	1. Programming Fundamentals using Python 2. Database Management Systems (DBMS) 3. Machine Learning Fundamentals 4. Data Analytics and Visualization Basics 5. Basic Understanding of AutoML and Artificial Intelligence Concepts	1. Design and develop an intelligent recommendation system for automated machine learning model selection and workflow generation. 2. Analyze dataset characteristics such as data type, dimensionality, feature distribution, and class imbalance to identify suitable machine learning approaches. 3. Implement recommendation algorithms that suggest appropriate models, preprocessing techniques, and AutoML workflows for predictive tasks. 4. Evaluate the effectiveness of the recommendation system in improving model selection accuracy, predictive performance, and machine learning development efficiency.
48	CSE-AI/ML	Mrs. Lalli	TPOT-Based Automated Pipeline Generator	AUTO ML	7CS 9281	7CSI281	This project develops an automated machine learning solution using TPOT to generate, optimize, and evaluate end-to-end machine learning pipelines. The system automates data preprocessing, feature selection, model selection, and hyperparameter tuning, reducing manual effort while improving model performance and development efficiency.	1. Programming Fundamentals using Python 2. Data Structures and Algorithms 3. Database Management Systems (DBMS) 4. Machine Learning Fundamentals 5. Data Analytics and Visualization Basics	1. Design and develop an automated machine learning framework using TPOT for end-to-end pipeline generation and optimization. 2. Implement automated data preprocessing, feature selection, model selection, and hyperparameter tuning techniques using genetic programming. 3. Evaluate and compare generated machine learning pipelines based on performance, efficiency, and predictive accuracy. 4. Develop a user-friendly interface for pipeline generation, model evaluation, and deployment support with minimal human intervention.
49	CSE-AI/ML	Dr M V Madan Kumar Kukumuri	End-to-End AutoML Web Application	AUTO ML	7CS 9281	7CSI281	This project develops a comprehensive AutoML web platform that automates data preprocessing, feature engineering, model selection, hyperparameter optimization, deployment, monitoring, and explainability. The system enables users to build, evaluate, deploy, and manage machine learning models through an intuitive web-based interface with minimal programming effort.	1. Programming Fundamentals using Python 2. Database Management Systems (DBMS) 3. Machine Learning Fundamentals 4. Web Application Development (Frontend and Backend Basics) 5. Data Analytics and Visualization Fundamentals	1. Design and develop a web-based AutoML platform that automates data preprocessing, feature engineering, model selection, and machine learning workflow management. 2. Implement automated model training, hyperparameter optimization, and performance evaluation techniques to generate high-quality predictive models. 3. Integrate Explainable AI (XAI), model deployment, and monitoring capabilities to support transparent and production-ready machine learning solutions. 4. Evaluate the effectiveness of the proposed platform in reducing development effort, improving model performance, and simplifying machine learning adoption for end users.
50	CSE-AI/ML	Dr. M Srinivas	Company- Adaptive AUTOML Frame work for Automated Model Selection and Optimization	AUTO ML	7CS 9281	7CSI281	Develop an adaptive AutoML framework that automatically analyzes datasets, selects suitable machine learning models, optimizes hyperparameters, and evaluates model performance. The system aims to reduce manual experimentation and identify the most efficient model based on accuracy, computational cost, and training time for different real-world applications.	1. Python Programming, 2. Fundamentals of Machine Learning 3. Data Preprocessing and Feature Engineering, 4. Model Evaluation and Hyperparameter Tuning	1. Develop an automated framework for selecting suitable machine learning models. 2. Apply hyperparameter optimization techniques to improve model performance. 3. Compare and evaluate multiple models using appropriate performance metrics.
51	CSE-AI/ML	Dr.Saswathi Debnath	Deep Learning-Based Automated Waste Classification and Smart Recycling System	IMAGE & VIDEO ANALYTICS	7CS 9248	7CSI248	Develop a deep learning-based intelligent system to automatically classify waste materials such as plastic, paper, metal, glass, and organic waste from images. The system supports efficient waste segregation and promotes sustainable recycling practices through automated decision-making.	1. Python Programming Fundamentals 2. Deep Learning and Computer Vision Basics 3. Image Processing Concepts	1. Develop a deep learning model for automated waste classification. 2. Accurately identify and categorize different types of recyclable waste. 3. Support smart recycling and waste management through AI-driven automation
52	CSE-AI/ML	Dr. Smitha Raj Gopal	Intelligent Retail Store Analytics Using Video Surveillance	IMAGE & VIDEO ANALYTICS	7CS 9248	7CSI248	Develop an AI-powered video analytics system that analyzes retail store surveillance footage to monitor customer movement, estimate footfall, identify shopping patterns, and generate actionable business insights for improving store operations and customer	1. Python Programming Fundamentals 2. Computer Vision and Deep Learning Basics 3. Image and Video Processing Concepts	1. Develop a video analytics system for customer behavior and footfall analysis. 2. Apply computer vision techniques to extract insights from surveillance videos. 3. Generate data-driven recommendations to enhance retail operations and customer engagement
53	CSE-AI/ML	Dr. Smitha Raj Gopal	Automated Road Damage Detection and Severity Assessment System	IMAGE & VIDEO ANALYTICS	7CS 9248	7CSI248	Develop a deep learning-based system to automatically detect road damages such as potholes, cracks, and surface defects from images or videos. The system assesses damage severity and assists authorities in prioritizing road maintenance and improving transportation safety	1. Python Programming Fundamentals 2. Deep Learning and Computer Vision Basics 3. Image Processing Concepts	1. Develop an automated system for road damage detection and classification. 2. Assess the severity of detected road defects using AI techniques. 3.Support efficient road maintenance planning through data-driven insights
54	CSE-AI/ML	Dr. Smitha Raj Gopal	Automated Sports Performance Analytics and Event Detection System	IMAGE & VIDEO ANALYTICS	7CS 9248	7CSI248	This project develops an AI-powered sports video analytics platform that detects players, tracks movements, recognizes game events, and evaluates player performance. The system generates tactical insights and performance metrics from sports videos, enabling coaches, analysts, and teams to make data-driven strategic decisions.	1. Programming Fundamentals using Python 2. Data Structures and Algorithms 3. Machine Learning and Artificial Intelligence Fundamentals 4. Computer Vision and Image Processing Basics 5. Data Analytics and Visualization Fundamentals	1. Design and develop a video analytics platform for automated player detection, tracking, and performance evaluation from sports videos. 2. Implement computer vision and deep learning techniques to identify players, track movements, and recognize key game events. 3. Generate tactical insights, player statistics, and performance metrics to support strategic decision-making in sports. 4. Develop interactive dashboards and visualization tools for analyzing player behavior, team performance, and match events.
55	CSE-AI/ML	Dr M V Madan Kumar Kukumuri	Real-Time Helmet Detection Using CCTV Video	IMAGE & VIDEO ANALYTICS	7CS 9248	7CSI248	Develop a real-time video analytics system to detect motorcycle riders and identify helmet usage from CCTV footage. The system applies computer vision and object detection techniques to recognize helmet and non-helmet cases and automatically identify road safety violations.	1.Python Programming, 2. Fundamentals of Image Processing, 3. Basics of Computer Vision, 4. Fundamentals of Machine Learning and Deep Learning	1.Develop a system to detect motorcycle riders from CCTV video. 2. Identify and classify helmet and non-helmet riders. 3. Apply object detection and video analytics techniques to real-time footage. 4. Generate automated alerts for detected helmet safety violations.
56	CSE-AI/ML	Dr.Saswathi Debnath	Real-Time Multi-Camera Crowd Behavior Analysis and Anomaly Detection	IMAGE & VIDEO ANALYTICS	7CS 9248	7CSI248	Develop an AI-powered video analytics system that processes multiple CCTV camera feeds to monitor crowd density, detect abnormal activities, identify suspicious behavior, and generate real-time alerts for enhanced public safety and security.	1. Python Programming Fundamentals 2. Computer Vision and Deep Learning Basics 3. Image and Video Processing Concepts	1. Develop a real-time multi-camera crowd monitoring and anomaly detection system. 2. Apply computer vision and deep learning techniques for behavior analysis. 3. Generate automated alerts for abnormal or suspicious crowd activities.
57	CSE-AI/ML	Dr.Saswathi Debnath	Development of a Travel-Related Software App for Mobile Phones to Capture Trip-Related Information	IMAGE & VIDEO ANALYTICS	7CS 9248	7CSI248	Develop a mobile application that enables users to capture, organize, and manage trip-related information such as locations, expenses, photos, videos, travel routes, and visited places. The app provides a centralized platform for efficient travel planning, tracking, and experience management.	1. Mobile Application Development Fundamentals 2. Basic Programming (Java/Kotlin/Flutter/React Native) 3. Database and API Integration Concepts	1. Develop a mobile application for capturing and managing travel-related information. 2. Integrate GPS, multimedia storage, and trip-tracking functionalities. 3. Generate organized travel records and insights to enhance user travel experiences.
58	CSE-CS	Dr. Nishanth Tripathi	Cryptocurrency transaction tracing tool using AI and Blockchain analytics.	Secure Access Edge	7CS 9291	7CSI291	This project develops an AI-powered cryptocurrency transaction tracing tool that analyzes blockchain transaction data to track the flow of digital assets, identify suspicious activities, and trace fund movements across wallets. The system supports fraud investigation, financial crime detection, and blockchain-based digital forensic analysis through intelligent transaction monitoring and visualization.	Programming Fundamentals using Python Data Structures and Algorithms Database Management Systems (DBMS) Computer Networks and Cyber Security Fundamentals Basic Understanding of Blockchain Technology, Cryptocurrencies, and Artificial Intelligence (AI)	Design and develop a blockchain-based cryptocurrency transaction tracing system for tracking digital asset movements across wallets. Apply artificial intelligence and graph analytics techniques to identify suspicious transaction patterns and potential fraudulent activities. Analyze blockchain transaction networks to trace fund flows and identify possible end recipients involved in financial crimes. Develop interactive dashboards and visualization tools for monitoring, investigating, and interpreting cryptocurrency transaction data. Evaluate the effectiveness of the proposed system in supporting digital forensics, fraud detection, and blockchain-based financial investigations.

59	CSE-CS	Dr. Rathnakar Achary	AI-Based Detection of Malicious Sideloaded Android Applications Using Runtime Threat Intelligence	Mobile Penetration Testing and Ethical Hacking	7CS 9292	7CS1292	This project develops an AI-based framework for detecting malicious sideloaded Android applications using runtime threat intelligence. By monitoring application behavior, system calls, network activities, and resource usage after installation, the system identifies emerging malware threats and protects users from data theft, unauthorized access, and malicious activities.	Programming Fundamentals using Python or Java Data Structures and Algorithms Computer Networks and Cyber Security Fundamentals Machine Learning and Artificial Intelligence Basics Mobile Application Development and Android Fundamentals	Design and develop a runtime threat intelligence framework for detecting malicious sideloaded Android applications. Collect and analyze application behavioral data, including system calls, permissions, network traffic, and resource utilization. Apply machine learning techniques to identify anomalous behaviors and classify potentially malicious applications. Develop a monitoring and alerting system for real-time detection of malware activities on Android devices. Evaluate the effectiveness of the proposed solution in detecting emerging mobile threats and enhancing Android device security.
60	CSE-CS	Dr. Rathnakar Achary	Android Privilege Escalation and Exploit Detection Framework Using Behavioral Analytics	Mobile Penetration Testing and Ethical Hacking	7CS 9292	7CS1292	This project develops a behavioral analytics framework for detecting Android privilege escalation and exploitation attempts in real time. By monitoring application activities, permission usage, system interactions, and abnormal behavioral patterns, the system identifies potential security breaches and prevents unauthorized access to sensitive device resources.	Programming Fundamentals using Python or Java Data Structures and Algorithms Operating Systems and Android Fundamentals Computer Networks and Cyber Security Fundamentals Machine Learning and Behavioral Analytics Basics	Design and develop a behavioral analytics framework for detecting privilege escalation and exploit attempts on Android devices. Monitor and analyze application behaviors, permission requests, system calls, and resource access patterns at runtime. Apply machine learning and anomaly detection techniques to identify suspicious activities indicative of exploitation attempts. Develop a real-time alerting and monitoring system to detect and mitigate privilege abuse before elevated access is obtained. Evaluate the effectiveness of the proposed framework in improving Android device security and detecting emerging exploitation techniques.
61	CSE-CC	Dr. Sathesh K	Smart Healthcare Management System with DevOps Automation	DevOps Automation and Optimization	7CS 9286	7CS 1286	This project develops a Smart Healthcare Management System integrated with DevOps automation to streamline patient management, appointment scheduling, electronic health records, and healthcare services. The system employs automated CI/CD pipelines, containerization, monitoring, and cloud deployment to ensure scalability, reliability, security, and continuous service delivery.	Programming Fundamentals using Python, Java, or JavaScript Database Management Systems (DBMS) Web Application Development Cloud Computing and DevOps Fundamentals Software Engineering and System Design	Design and develop a smart healthcare management platform for patient records, appointments, billing, and healthcare service management. Implement DevOps practices including Continuous Integration and Continuous Deployment (CI/CD) for automated software delivery. Deploy healthcare services using containerization and orchestration technologies to improve scalability and reliability. Integrate monitoring, logging, and automated testing mechanisms to ensure system performance, security, and availability. Evaluate the effectiveness of DevOps automation in improving operational efficiency, deployment speed, and healthcare service quality.
62	CSE-CC	Dr. Chetan Shelke	AI-Driven Autoscaling Using Terraform	DevOps Automation and Optimization	7CS 9286	7CS 1286	This project develops an AI-driven autoscaling framework that dynamically provisions and manages cloud resources using Terraform. By analyzing workload patterns and system metrics, the platform predicts resource requirements and automatically scales infrastructure to optimize performance, availability, and cloud resource utilization.	Programming Fundamentals using Python Cloud Computing Fundamentals DevOps and Infrastructure as Code (IaC) Concepts Machine Learning and Data Analytics Basics Linux, Networking, and System Administration Fundamentals	Design and develop an AI-driven autoscaling framework for dynamic cloud resource management. Implement machine learning models to analyze workload patterns and predict future infrastructure demands. Automate cloud infrastructure provisioning and scaling using Terraform and Infrastructure as Code (IaC) principles. Integrate monitoring and performance analytics to support intelligent scaling decisions in real time. Evaluate the effectiveness of AI-driven autoscaling in improving resource utilization, system performance, cost optimization, and service availability.
63	ASAE-ECE	Srikanth Iapu	Glass and Optical semiconductor packaging for Quantum Hardware	Semiconductor Packaging and Manufacturing	7EC 9317	7EC1317	Glass and optical semiconductor packaging focuses on integrating photonic and electronic components using glass-based substrates and advanced encapsulation techniques. Glass offers excellent optical transparency, low dielectric loss, and high thermal and chemical stability, making it ideal for high-speed data transmission and photonic integrated circuits. This packaging approach enables efficient coupling of light sources, waveguides, and detectors while maintaining signal integrity. It is widely used in applications such as fiber-optic communication, LiDAR, and optical sensors. Emerging trends include wafer-level glass packaging, 3D integration, and hybrid photonic-electronic systems for next-generation computing and communication technologies.	Understanding of Semiconductors, devices, and basic level of packaging.	The project can produce interdisciplinary paper on wafer-level glass packaging, thermal management, and reliability analysis under extreme conditions relevant to telecom and defense. Conference publications in venues like IEEE Electronic Components and Technology Conference and SPIE Photonics West will enhance global visibility.
64	ASAE-EEE	Dr. Atmanadamaya	IoT based Renewable Energy Monitoring System for Microgrids	Power System Operation and Control	7EE 9015	7EE 1015	Develop an IoT-based monitoring system for solar or wind microgrids in rural areas, providing real-time data on energy generation, storage, and consumption. The system should alert users to inefficiencies or maintenance needs.	1. IoT sensors 2. MATLAB Simulink 3. Microgrid Setup	A hardware-software prototype improving microgrid efficiency by 15%, with a mobile app for community operators to manage energy distribution.
65	ASAE-EEE	Dr. Atmanadamaya	Develop a power system state estimation for renewable energy based network using AI	Power System Operation and Control	7EE 9015	7EE 1015	Develop a power system state estimation technique for a real PV and Diesel generation based modern power system network using machine learning classification.	1. MATLAB Simulink 2. MATLAB 2026 Licensed version	A software prototype for improvement of power system state estimation with more efficiency than conventional methods which can be computable for smart grid network also.
66	ASAE-EEE	Dr. Swetha Shekarappa G	AI-Based Adaptive Protection Scheme for Renewable Integrated Microgrids	Power System Protection and Switchgear	7EE 9014	7EE1014	This project develops an AI-based adaptive protection scheme for microgrids with renewable energy sources. It focuses on intelligent fault detection and relay coordination under varying operating conditions, addressing challenges like bidirectional power flow and dynamic fault levels in modern grids.	1. Basics of Power System Protection 2. Fundamentals of Machine Learning 3. Knowledge of Microgrids and Renewable Energy Systems 4. MATLAB/Python simulation skills	1. Develop an AI-based fault detection model 2. Analyze protection performance under dynamic conditions 3. Compare conventional and intelligent protection schemes 4. Improve relay coordination using adaptive techniques 5. Publish results in a journal focusing on smart grid protection
67	ASAE-EEE	Dr. Susmita Adhikary	Hybrid SVC- STATCOM Architecture with Mode Switching	Flexible AC Transmission Systems	7EE 9212	7EE1212	The idea is to design a hybrid SVC- STATCOM system that controls voltage in a power system. It automatically switches between two methods based on grid conditions to keep voltage stable. The system is tested in MATLAB/Simulink under different situations like load changes and faults.	1. Power Electronics 2. FACTS Coding/Simulation 3. MATLAB	1. Conference Publication (Scopus Indexed) 2. Journal Publication
68	ASAE-CE	Dr. Mithun Hanumesh	Sustainability and affordability: A case study of making an underprivileged site sustainable in a cost-effective way	Sustainable Construction Materials	7CE 9019	7CE1019	Develop a practical low-cost plan to improve sustainability in an underprivileged site through water, waste, energy and greening interventions. Assess needs, estimate budget and recommend affordable solutions with measurable social and environmental benefits.	1. Basics of sustainability 2. Cost estimation fundamentals 3. Site survey skills	1. Conduct baseline site assessment. 2. Propose low-cost sustainable interventions. 3. Prepare budget and implementation plan. 4. Evaluate expected social and environmental impact.
69	ASAE-AE	Dr. Raja Munusamy	Design and Simulation of Hybrid Aircraft Navigation System using INS-GPS Integration	Aircraft Navigation Systems	A7AE 9244	A7AE 1244	This project focuses on designing and simulating a hybrid aircraft navigation system by integrating inertial navigation (INS) and GPS. It emphasizes sensor fusion, error correction, and real-time positioning accuracy using MATLAB/Simulink for aerospace applications	1. Fundamentals of Aircraft Navigation 2. Knowledge of Sensors (Gyroscope, Accelerometer) 3. Basic MATLAB/Simulink skills 4. Understanding of coordinate systems and transformations	1. Design a hybrid navigation model using sensor fusion methods 2. Analyze and model INS and GPS navigation systems 3. Simulate aircraft position and velocity estimation 4. validate navigation errors and implement correction techniques
70	ASAE-AE	Dr. Rohith I J	Design and Implementation of AI based Architecture for Autonomous Fault Management in CubeSats	Spacecraft Design	7AE 9206	7AE 1206	This project involves designing an onboard AI based architecture for CubeSats of any configuration. Students will develop a lightweight AI system capable of autonomous fault detection and recovery to ensure mission success during deep-space latency. The work includes synthetic data training and hardware-in-the-loop validation.	1. Basic Spacecraft Engineering: Familiarity with CubeSat subsystems (ADCS, EPS, COMM) and mission operations. 2. Basics of C++ 3. Embedded Systems Knowledge: Experience with low-power hardware like ARM Cortex-M or Raspberry Pi modules.	"1. Verified AI-FDIR Software Simulation Environment 2. Validated Hardware-in-the-Loop (HIL) Prototype; 3. Patent / Journal Publication"
71	ASAE-AE	Dr. Deepak Sampathkumar	Self-Healing Carbon Fiber Composites Design for Damage-Tolerant Aircraft Structures	Aircraft Design	7AE 9204	7AE 1204	Design and develop self-healing carbon fiber reinforced composites for aircraft structures capable of autonomously repairing micro-cracks and restoring mechanical strength. The study integrates material design, structural analysis, and experimental validation to enhance damage tolerance, durability, and safety of next-generation aerospace airframes design.	1. Aerodynamics, 2. Aircraft Structures, 3. Strength of Materials, 4. Composite Materials, 5. Aircraft Design, and 6. Materials Science	1. Journal Publication, 2. Conference presentation

